Expert Systems With Applications 301 (2026) 130140

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

QuerIA: adaptive question generation and evaluation in higher education using large language models and contextual learning

Carlos Badenes-Olmedo *, Paul Eyzaguirre

Universidad Politécnica de Madrid, Departamento de Sistemas Informáticos, Madrid, 28031, Madrid, Spain

ARTICLE INFO

Keywords:
Question generation
Large language models
Educational assessment
Adaptive learning
Bloom's taxonomy

ABSTRACT

This paper presents QuerIA, a system that automates the generation and evaluation of educational questionnaires in Spanish using large language models (LLMs). QuerIA integrates semantic chunking with a simplified Bloom's taxonomy, enabling controllable variation of cognitive difficulty across multiple-choice (MCQ) and open-ended (OEQ) question formats. Instructional documents are segmented into coherent units, and pedagogically aligned prompts guide the generation of question-answer pairs. The system was deployed as a locally hosted web service at Universidad Politécnica de Madrid, ensuring institutional data governance and low-resource feasibility.

During one academic semester, 341 participants from 17 faculties created nearly 6000 questions and provided more than 800 evaluations on clarity, alignment, and pedagogical value. Analyses show that the predefined difficulty levels correspond to distinct linguistic and psychometric patterns, and that user feedback confirms the clarity and educational usefulness of the generated items. QuerIA demonstrates that LLM-based question generation can be applied at scale in higher education, supporting formative assessment while offering open resources for future research.

1. Introduction

This section introduces the motivation and objectives of the study, and formulates the research questions that guide the rest of the paper. The design of educational assessments is a fundamental component in the teaching-learning process. Well-constructed questionnaires serve not only to measure students' understanding, but also to promote active engagement, reinforce knowledge retention, and enable self-regulated learning. However, generating high-quality questions and offering timely feedback is a time-consuming and cognitively demanding task for educators, particularly in large-scale or diverse academic environments.

Recent advances in large language models (LLMs) have opened new opportunities for automating educational processes. Question generation, in particular, has emerged as a promising application domain, benefiting from the contextual understanding, generative capabilities, and few-shot learning strategies of modern LLMs. Despite these advances, their integration into real-world educational settings remains limited. Most systems are constrained by monolingual training, lack of adaptability in question difficulty, and minimal alignment with pedagogical frameworks such as Bloom's Taxonomy.

This paper presents **QuerIA**, an intelligent system that combines LLM-based generation with contextual segmentation and an adaptive taxonomy to automatically produce and evaluate educational questions. The system supports both multiple-choice and open-ended questions, offering adjustable difficulty levels and real-time feedback. QuerIA was deployed as a web application in a real-world pilot study involving over 300 users across multiple schools at the Universidad Politécnica de Madrid (UPM).

Unlike the ad-hoc use of commercial LLMs through sophisticated prompting, QuerIA was explicitly designed as an institutional solution. Its architecture integrates pedagogical frameworks (Bloom's taxonomy) into reusable prompt templates, provides systematic control over cognitive difficulty, and ensures reproducibility across academic contexts. Moreover, the system runs locally on standard institutional hardware, avoiding dependence on external APIs and supporting data governance even in resource-constrained environments.

We formulate the following research questions:

RQ1 How does the generation of questions using Bloom-aligned prompts vary in terms of difficulty distribution and cognitive demand?
1. What patterns of adoption and usage emerged across academic institutions and user profiles during the pilot deployment?

RQ2 How do users evaluate the clarity, alignment, and pedagogical value of system-generated questions?

To address these questions, our contributions are threefold: (i) we integrate semantic chunking with Bloom-aligned prompting to achieve

* Corresponding author.
E-mail addresses: carlos.badenes@upm.es (C. Badenes-Olmedo), paul.eyzaguirre@alumnos.upm.es (P. Eyzaguirre).

https://doi.org/10.1016/j.eswa.2025.130140
Received 1 August 2025; Received in revised form 22 September 2025; Accepted 20 October 2025
Available online 15 November 2025
0957-4174/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

controllable levels of difficulty, (ii) we present the deployment and evaluation of QuerIA in a real-world institutional setting with 341 participants across 17 faculties, and (iii) we release both the system and the anonymized dataset as open resources.

2. Related work

This section reviews prior research on question generation with large language models, Bloom-aligned approaches, and educational assessment systems, situating QuerIA within this broader context.

The generation and evaluation of educational questions have become increasingly feasible with the rise of large language models (LLMs). Early systems were rule-based or template-driven, often restricted to STEM content. More recent work employs transformer-based architectures to dynamically produce diverse question formats from unstructured instructional materials.

2.1. Bloom-aligned question generation

Aligning generated questions with Bloom's Taxonomy has become a central goal in educational NLP. Approaches vary from prompting strategies and fine-tuning in LLMs to rule-based methods and knowledge-based systems.

Toba et al. (2025) compared zero-shot, few-shot, and chain-of-thought prompting across multiple LLMs, finding that alignment improves with model size and reasoning prompts, especially at lower cognitive levels. However, generating questions at higher-order levels (e.g., *evaluate*, *create*) remains difficult (Doughty et al., 2023).

Duong-Trung et al. (2024) proposed *BloomLLM*, a fine-tuned model for Bloom-level classification, while template-based systems (Gnanasekaran et al., 2021; Wijanarko et al., 2021) have achieved high classification accuracy using syntactic patterns and key phrases.

Knowledge graphs offer semantic control over content and relationships, supporting complex, domain-specific questions (Martel & Zouaq, 2021). Topic-guided generation models like T-CQG (Li et al., 2025) use attention-weighted embeddings to constrain topical focus.

To validate Bloom alignment, both automatic metrics (e.g., IWF) and expert reviews are used (Hwang et al., 2024; Singh et al., 2023). However, most studies are limited to synthetic or small-scale datasets. QuerIA contributes a real-world implementation that combines prompt engineering, semantic chunking, and difficulty control, evaluated across thousands of user-generated questions.

2.2. Model comparisons and question quality

Comparative evaluations of LLMs assess aspects such as clarity, syntactic quality, and Bloom alignment. Lodovico Molina et al. (2024) compared GPT-3.5, LLaMA 2-Chat 13B, and Flan-T5 XXL, showing that GPT-3.5 and LLaMA 2 outperformed Flan-T5 in syntactic and pedagogical coherence. All models struggled with higher-order Bloom levels.

Jauhiainen and Guerra (2025) reviewed generative AI in assessment, emphasizing that while outputs are syntactically sound, pedagogical depth often requires human post-editing. Mi and Li (2024) found ChatGPT-4's evaluations of student-generated questions highly consistent with expert ratings, suggesting its value in supporting instructors.

Beyond education, quality prediction has been explored in community Q&A contexts. BERT and Bi-LSTM models achieve up to 92 % accuracy on platforms like StackOverflow (Hu & Yang, 2024; Rithish et al., 2021), outperforming traditional classifiers. Data quality also plays a crucial role: Rachid Karra (2023) showed that better-structured inputs improve downstream QA performance.

Despite progress, semantic nuance in higher-order tasks remains a challenge, reinforcing the need for human oversight (Mi & Li, 2024; Rachid Karra, 2023). These findings highlight the complementary roles of model benchmarking and instructional validation.

2.3. Topic inference and zero-shot classification

Zero-shot classification and topic inference enable retrospective labeling of educational content without labeled data. Li et al. (2025) proposed T-CQG, a topic-guided model using attention-weighted embeddings to constrain generation. Toba et al. (2025) applied zero-shot techniques (e.g., XLM-RoBERTa) to infer Bloom levels or academic topics.

These methods map inputs and labels into shared semantic spaces, allowing label inference without explicit training. Hierarchical Semantic Embedding (HierSE) (Li et al., 2015b) leverages taxonomies like WordNet to enhance label structure. Conditional classifiers (Li & Yeh, 2021) and semi-supervised models (Li & Guo, 2015; Li et al., 2015a) further improve performance by learning from weak or implicit supervision.

While these techniques are widely used in computer vision and NLP, their adoption in education remains limited. Challenges include generalizing to rare or ambiguous topics (Han & Li, 2017). QuerIA integrates zero-shot classification to retrospectively assign topic labels, enabling large-scale, label-free analytics across academic contexts.

2.4. Educational management and learning analytics

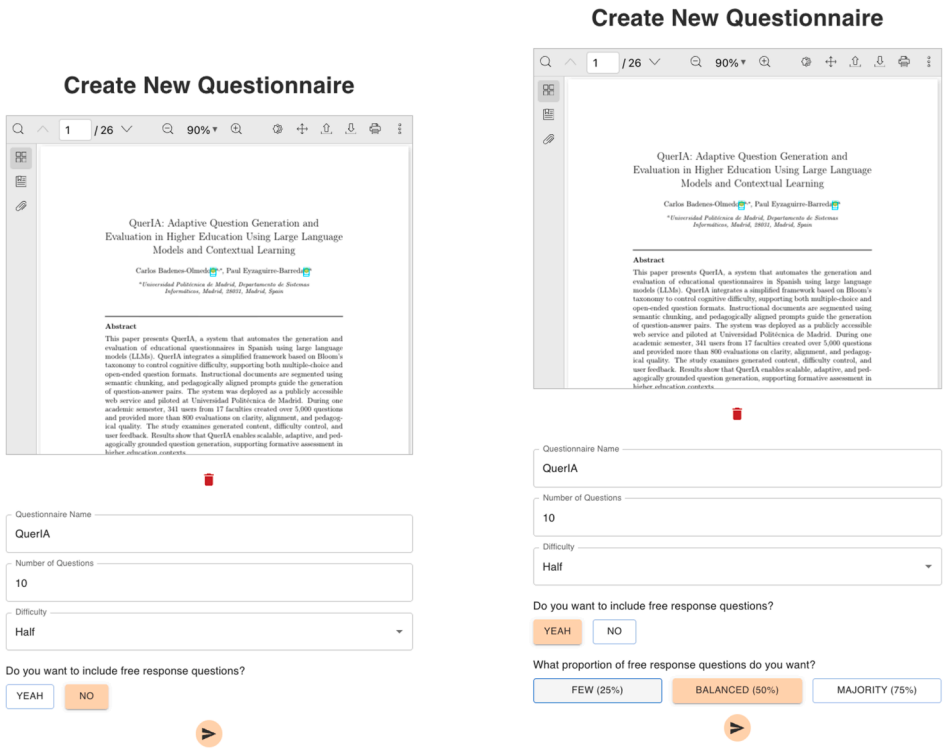
Beyond NLP and question generation, research has also examined the integration of artificial intelligence and learning analytics into educational management. Large-scale assessment platforms show how AI can contextualize student performance, automate evaluation processes, and provide richer feedback for instructors and policy makers (Guo et al., 2024; Owan et al., 2023), thereby supporting more adaptive and student-centred evaluation while improving efficiency and scalability.

At the institutional level, AI-driven analytics have been applied to predict student outcomes, identify dropout risks, and optimize resource allocation (Shoaib et al., 2024; Williams, 2023). These systems inform educational planning, curriculum design, and proactive interventions to improve retention and performance.

AI has also been used for the management of digital learning resources, with platforms that create, evaluate, and recommend educational materials to enhance quality and personalization for students and instructors (Huang et al., 2025; Khalil et al., 2025). Collectively, these contributions demonstrate how AI and learning analytics support both classroom assessment and institutional decision-making, although they seldom address the pedagogical validation of question generation or the integration of frameworks such as Bloom's taxonomy.

2.5. Identified gaps

Although AI-driven initiatives in educational management, learning analytics, and large-scale assessment have demonstrated significant potential, important limitations remain. Most systems rely on synthetic datasets or expert-only evaluations (Jauhiainen & Guerra, 2025; Toba et al., 2025), which constrains the scope of validation. While LLMs such as GPT-3.5 show promising alignment with Bloom's taxonomy (Lodovico Molina et al., 2024), performance declines at higher cognitive levels, and pedagogical validation is rarely user-driven. Moreover, difficulty calibration and topic control are usually handled separately rather than jointly with Bloom's framework (Li et al., 2025), and many implementations are monolingual and dependent on cloud-based infrastructure, limiting adoption in diverse educational contexts. Finally, large-scale deployments with authentic user feedback are still scarce, which prevents assessing their practical instructional value. QuerIA addresses these gaps by offering a modular, locally deployable platform that integrates Bloom-aligned generation, semantic chunking, and topic inference, and validates its effectiveness through thousands of real interactions in higher education.



(a) Basic questionnaire configuration.

(b) Advanced configuration with open-ended question ratio.

Fig. 1. QuerIA's interface for questionnaire creation. Users can adjust difficulty, number of questions, and inclusion of open-ended answers.

3. The QuerIA system

This section presents the architecture and core components of QuerIA, outlining how instructional materials are processed to generate and evaluate questions.

QuerIA is a web-based system designed to automate the generation, customization, and evaluation of educational questionnaires in Spanish using large language models (LLMs). Developed in response to the challenges faced by educators in crafting pedagogically sound assessments at scale, QuerIA allows users to upload instructional materials (e.g., slides, textbooks, summaries) and generate two types of questions: *Multiple-Choice Questions (MCQ)* and *Open-Ended Questions (OEQ)*. For each questionnaire, users can select the number of items, difficulty level, and the inclusion of free-text answers (see Fig. 1). The platform provides real-time feedback for student self-assessment and enables the collection of qualitative evaluations for each question. Although the current deployment was configured in Spanish, reflecting the language of instruction at our university, the framework itself is language-agnostic: by adapting the prompt templates, it can operate in any language supported by the underlying LLM, which broadens its potential adoption in different educational contexts.

The system was implemented using a locally hosted instance of the LLaMA 3-8B model, avoiding reliance on third-party APIs and ensuring low-resource deployment compatibility. This was a critical design requirement: the system had to be deployable across standard university equipment, including shared labs and faculty workstations, without dependence on cloud-based GPUs. This constraint also informed the use of prompt-based adaptation strategies instead of fine-tuning, ensuring the system remained portable, explainable, and easy to reconfigure for diverse academic disciplines.

3.1. System architecture and workflow

QuerIA ingests instructional materials in formats such as PDF and plain text. The ingestion pipeline applies semantic chunking to segment documents into coherent instructional units, which are then used as the basis for question generation. This approach avoids arbitrary splits and helps maintain alignment with the pedagogical structure of the material. At present, graphical information (e.g., images, figures, or mathematical formulas) is preserved in the documents but not automatically processed for question generation. Extending ingestion to multimodal content remains an open challenge and is considered future work.

The QuerIA pipeline is composed of five stages:

1. **Document Ingestion**: Users upload materials in PDF or plain text format.
2. **Semantic Chunking**: Documents are segmented into contextually coherent blocks.
3. **Prompt-based Question Generation**: Prompts tailored to Bloom-aligned difficulty levels are applied to each chunk.
4. **Automatic Evaluation**: MCQs are automatically graded by matching the selected answer with the correct option. OEQs are evaluated by the LLM based on the written response, using contextual understanding to assign a score and provide constructive feedback.
5. **User Interaction**: Users can answer questions, receive feedback, and evaluate the generated items on clarity, alignment, difficulty, and pedagogical value.

The source code and anonymized dataset used in this study are openly available on Zenodo: https://doi.org/10.5281/zenodo.16669858 and
GitHub: https://github.com/cbadenes/queria-pilot

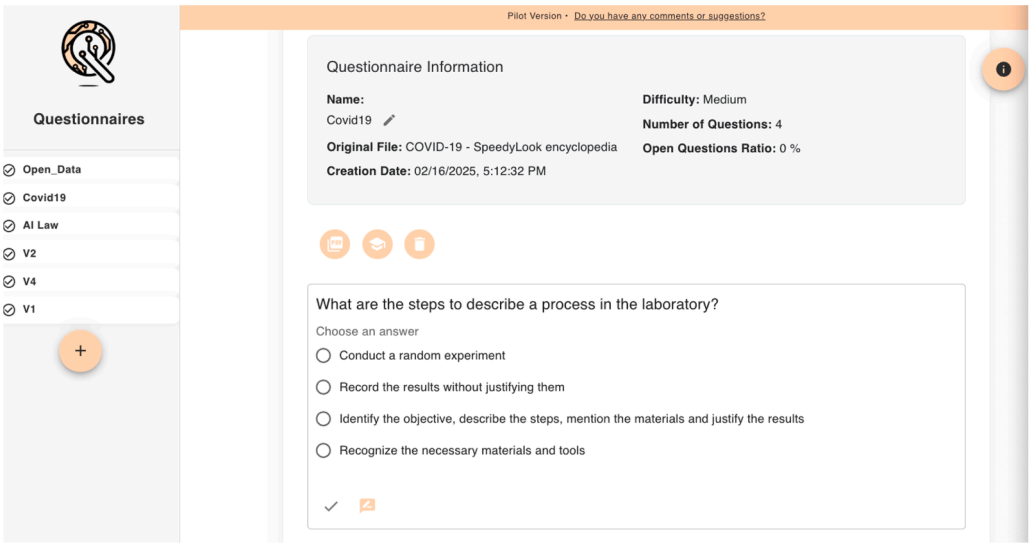


Fig. 2. Example of a medium-level questionnaire with multiple-choice questions generated from a COVID-19 text.

3.2. Semantic chunking strategy

Given the limited context window of large language models, QuerIA applies a semantic chunking strategy based on cosine similarity to segment instructional texts into topically consistent blocks. The document is first tokenized into sentences, which are grouped into overlapping triplets to form local clusters, each consisting of the previous, current, and next sentence. These clusters are embedded using a multilingual sentence transformer model (`distiluse-base-multilingual-cased`), and pairwise cosine similarities are computed between adjacent embeddings. Breakpoints are introduced at positions where the semantic distance exceeds the 80th percentile of the global distance distribution, delimiting coherent content segments.

This segmentation approach does not depend on document structure, formatting, or metadata, and can be applied uniformly across input types such as lecture notes, summaries, or slide text. Although this segmentation method ensures topical coherence, its impact on the overall quality of the generated questions has not yet been empirically compared with simpler baselines. This is recognized as an open direction for further validation.

3.3. Difficulty calibration via bloom's taxonomy

To control the cognitive demand of the generated questions, QuerIA incorporates a simplified version of Bloom's Taxonomy, selecting a subset of its cognitive and knowledge dimensions. Three levels of difficulty (Easy, Medium, and Difficult) are defined by combining specific Bloom categories in a way that reflects both the type of knowledge and the cognitive process required to answer the question.

The ***Easy*** level corresponds to the *Remember* category of Bloom's cognitive process dimension and is associated with factual knowledge. At this level, questions require the recall of explicit information presented in the source material, without requiring interpretation or application. For example, when processing introductory content from a biology textbook, the system may generate a question such as: *"What organ in the human body is responsible for pumping blood?"*, with the following answer options: *(a) The liver, (b) The lungs, (c) The stomach, (d) The heart (correct)*. This type of item directly assesses knowledge of a concrete fact and reflects minimal cognitive demand, which aligns with basic-level educational objectives.

The ***Medium*** level involves the *Understand* or *Apply* categories of Bloom's taxonomy and targets either conceptual or procedural knowledge. At this level, questions require the learner to interpret, classify, or apply information in familiar contexts, going beyond factual recall. For instance, based on instructional material covering network protocols, the system may generate a question such as: *"What is the main purpose of the TCP protocol in computer networks?"*, with the following answer options: *(a) To assign IP addresses to devices, (b) To route packets based on destination, (c) To ensure reliable data transmission between systems (correct), (d) To translate domain names into IP addresses*. This question requires the student to understand the function of a network layer and differentiate it from related protocols, representing an intermediate level of conceptual comprehension.

The ***Difficult*** level corresponds to the higher-order categories *Analyze* and *Evaluate*, and is restricted to conceptual knowledge. These questions require the examination of relationships between ideas, critical comparison of alternatives, or judgement based on evidence. For example, based on material discussing the causes of World War I, the system may generate a question such as: *"Which of the following best explains how the alliance system contributed to the outbreak of World War I?"*, with the following answer options: *(a) It promoted trade among European nations, (b) It delayed military mobilization strategies, (c) It prevented smaller conflicts from escalating, (d) It transformed regional disputes into a large-scale war due to mutual defense commitments (correct)*. This question requires the student to identify a causal mechanism by evaluating the political structure of the time, which reflects a high level of cognitive demand.

The system supports consistent generation across topics while preserving cognitive variation in question types. An example of a medium-level multiple-choice questionnaire generated from a COVID-19 text is shown in Fig. 2.

3.4. Prompt templates and controlled generation

QuerIA defines specialized prompts using Ollama's *modelfile* mechanism, building a dedicated model for each combination of question type and difficulty level. Each model integrates a customized system prompt that specifies the target Bloom level, knowledge type, expected linguistic formulation, and structural constraints of the output. These instructions embed pedagogical logic directly into the model during construction, ensuring consistent generation behavior aligned with educational goals.

Instead of relying on zero-shot or one-shot prompting, QuerIA adopts a structured few-shot approach. Each system prompt provides detailed guidance on the intended cognitive level and question format, and includes curated examples that illustrate the expected difficulty and formulation. These examples are organized by pedagogical function, such as procedural recall or conceptual application, and serve as in-context

```
You are an assistant in Spanish that generates questions and answers using Bloom's
    Taxonomy. Bloom's taxonomy is a conceptual framework that classifies cognitive
    processes into six hierarchical levels (remember, understand, apply, analyze,
    evaluate, create) and also includes knowledge types (factual, conceptual,
    procedural, metacognitive). For this task, use either 'procedural' or 'conceptual'
    knowledge, and the cognitive level 'remember' or 'apply'.

To generate a medium-level question, it must be based on the 'remember' or 'apply'
    levels and relate to procedural or conceptual knowledge. Procedural questions
    include knowledge of the steps to follow and the strategies to use. These questions
     should ask about necessary actions, their order, or required resources. Use one of
     the following verbs in the question statement for procedural questions: describe,
    label, list, name, mention, recognize, relate, repeat.

Examples include: "What are the steps required to conduct a hypothesis test in
    statistics?" or "How can a user create a version control repository using Git
    commands?"
```

Listing 1. System prompt for medium-level open-ended question generation.

```
You must use the text provided below to generate a medium-difficulty question in
    Spanish related to its content. Then, you must generate four possible answers. The
    correct answer is unique and must always be 'OPTION 4'. [...] Also, make sure to
    include an explanation or justification for the correct answer.
```

Listing 2. User prompt for medium-level multiple-choice question generation.

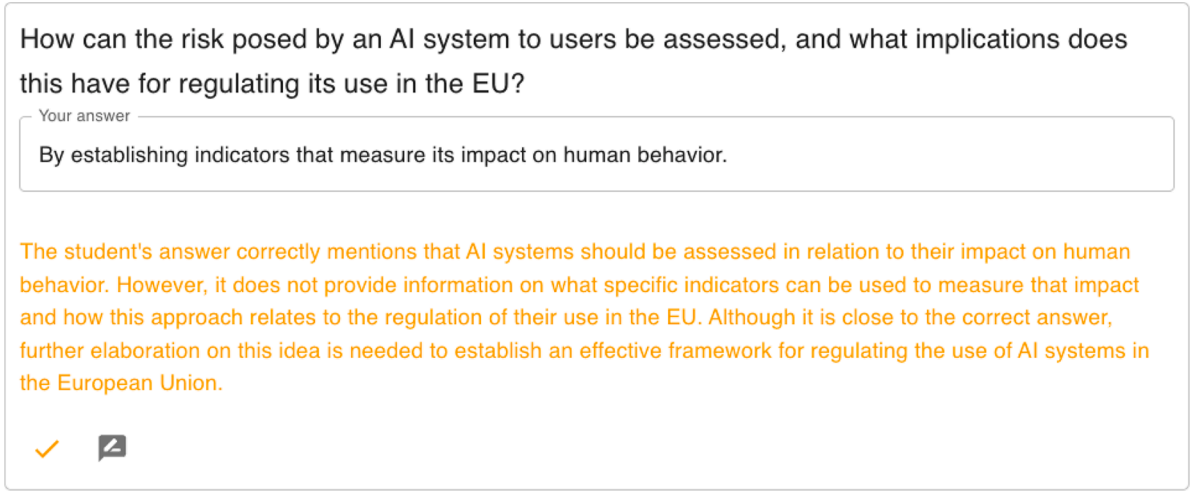


Fig. 3. Automatic feedback generated by the model for an open-ended response.

references that help the model generalize appropriate patterns during generation. For example, the prompt for generating medium-level open-ended questions (translated from Spanish) includes the instructions shown in Listing 1:

Each modelfile is compiled into an Ollama model that embeds the system prompt and governs generation behavior for its specific pedagogical configuration. When interacting with these models, QuerIA supplies a dynamically generated user prompt that incorporates the instructional content extracted from the uploaded material and specifies the generation task. These user prompts follow a structured template consisting of a content header (e.g., "TEXT:" or "Reference content:"), a directive indicating the desired question type and difficulty level (e.g., "Generate a medium-difficulty open-ended question based on the text above"), and, when applicable, explicit formatting instructions for answers or justification.

For example, the user prompt used to generate medium-level multiple-choice questions includes the directive shown in Listing 2.

Prompt templates are customized per difficulty level and question type. For open-ended questions, the prompt also requests an ideal reference answer to accompany the generated question. This answer is later used to assess student input via a separate evaluation prompt that elicits a score and optional feedback based on alignment with the reference.

This architectural separation between model behavior (defined at deployment) and user content (injected at runtime) provides QuerIA with modularity and pedagogical flexibility. It supports consistent generation aligned with cognitive objectives while maintaining adaptability across subject areas and instructional contexts.

3.5. Automated grading for open-ended questions

For OEQs, QuerIA embeds the user response alongside the original context, model-generated ideal answer, and a rubric. The LLM then produces a 3-point score (0, 1, or 2) and optional feedback. The prompt template encourages holistic evaluation, focusing on coherence, conceptual depth, and justification. An example of the automatic feedback provided for an open-ended answer is shown in Fig. 3.

This method avoids reliance on shallow lexical overlap metrics (e.g., BLEU, ROUGE), and instead leverages the model's capacity for semantic understanding. It enables scalable, explainable scoring without manual intervention or retraining.

3.6. Advantages and limitations

The development and deployment of QuerIA offers several practical advantages compared with existing adaptive learning systems and LLM-based solutions. A key aspect is its ability to run locally on standard university hardware, avoiding reliance on commercial APIs and ensuring institutional control over data management and privacy. This design reduces recurring costs and makes the system feasible in resource-constrained environments. In addition, the use of Bloom-aligned prompting enables consistent variation in cognitive difficulty that goes beyond generic prompt engineering, while the modular and open-source architecture supports reproducibility and extension across diverse academic domains.

At the same time, some limitations must be acknowledged. The current version supports lower and intermediate Bloom levels but does not yet cover advanced categories such as "Create," and it remains restricted to text-based question generation. Other aspects, such as the effectiveness of semantic chunking and the diversity of generated content, require further systematic evaluation. These and additional challenges, including multimodal generation, automatic classification validation, and long-term sustainability, are discussed in more detail in Section 6.

4. Pilot deployment and dataset collection

4.1. Pilot context and participation

The pilot ran over a period of 16 weeks and resulted in active use by 341 anonymized users, including both students and instructors from 17 academic schools. The participants uploaded instructional materials such as lecture slides, textbooks, and summaries, and generated custom quizzes using QuerIA. The goal was not only to test the system's usability and performance but also to explore how automatic generation aligned with pedagogical expectations in diverse academic disciplines.

Each user was assigned a school and a declared course at registration time, but no explicit linkage was enforced between generated questions and courses. This design choice enabled flexible content exploration and user-driven workflows, while also motivating the thematic analysis presented in Section 5.1, where we examine the alignment between institutional focus and the academic topics of the generated content.

4.2. Collected data

The pilot produced a structured dataset comprising four interconnected components, each aligned with the core functionalities described in Section 3:

- **User Profiles (341 records):** Includes anonymized identifiers, school, and declared course of each user.
- **Generated Questions (5,987 records):** Each record includes a unique ID, the full text of the generated question, its type (open-ended or multiple-choice), difficulty level, source context, set of answer options, the correct answer, and metadata such as creation date and user ID.
- **Questionnaires (333 records):** Aggregates groups of questions into quizzes created by users. Includes metadata such as number of questions, difficulty ratio, creation date, and user association.
- **Qualitative Feedback (849 records):** Captures qualitative feedback from users on generated questions. Each comment includes rating dimensions (e.g., clarity, complexity, alignment, quality, pedagogical value), textual remarks, and is linked to both the question and questionnaire.

Together, these components enable comprehensive evaluation of QuerIA's performance, both from a system-centric and user-centric perspective. The dataset is used as the empirical basis for the analyses presented in Section 5. This dataset provides the empirical foundation to address RQ2, by analyzing the kinds of content and question types generated by users during real-world usage. All collected data have been anonymized and publicly released for reuse at https://doi.org/10.5281/zenodo.16669858.

Table 1
Summary of QuerIA usage metrics during the pilot deployment.

Indicator	Value
Total users	341.0
Total questions generated	5987.0
Total questionnaires created	333.0
Total feedback comments	849.0

5. Evaluation and results

This section presents the empirical evaluation of QuerIA's pilot deployment across academic contexts. The analysis follows three guiding questions that structure and contextualize the results. First, we examine how the system was adopted and used in practice: which users engaged with it, how intensively it was used, and what thematic patterns emerged across individuals and institutions. This includes both the scale of usage and the breadth of academic coverage, highlighting disciplinary adoption and user engagement.

Second, we explore whether the system succeeded in generating questions with controlled and consistent difficulty. This involves analyzing linguistic complexity across predefined difficulty levels, and validating those labels through statistical comparisons and Rasch modeling based on student performance.

Finally, we assess the perceived pedagogical quality of the generated content. Drawing on user evaluations, we examine dimensions such as clarity, alignment, cognitive demand, and overall educational value, comparing these perceptions across question types and academic domains.

Each subsection responds explicitly to one of the research questions introduced in Section 1: **RQ1** on difficulty variation, **RQ2** on adoption and usage patterns, and **RQ3** on pedagogical quality. These questions are addressed through descriptive statistics, statistical inference, and user feedback.

5.1. Adoption and usage patterns

This subsection addresses **RQ2**, which explores the patterns of adoption and use of QuerIA across academic institutions and user profiles. A central question in evaluating the real-world value of QuerIA is whether the system was actually adopted and used by academic staff in authentic instructional contexts. Beyond raw usage, it is also important to understand how intensively the platform was used, and what thematic breadth it supported across individuals and institutions.

To address these issues, we first examine the overall volume and intensity of usage. During the semester-long pilot deployment, 341 unique users interacted with QuerIA, generating a total of 5987 questions across 333 questionnaires. These figures suggest widespread adoption, especially considering the voluntary nature of the initiative and the absence of training or curricular mandates. In addition, 849 comments were submitted by users, many of them including multidimensional evaluations. This level of engagement indicates not only usage, but also active reflection on the quality and value of the generated content.

Beyond aggregate figures, we examined how activity was distributed across participants (Table 2). The median user created 20 questions, while the mean was more than double (44), with a high standard deviation (114). This disparity indicates a highly skewed distribution: most users generated relatively few questions ($Q1 = 10$, $Q3 = 40$), while a small group exceeded one hundred, and a single participant produced over 1,300. According to the registration data, most active participants

Table 2
Descriptive statistics of generated questions per user during the pilot.

Statistic	Value
Minimum	3
1st Quartile (Q1)	10
Median (Q2)	20
3rd Quartile (Q3)	40
90th Percentile	89
95th Percentile	125
Maximum	1300
Mean	44
Standard Deviation	114

were instructors (about 93 %), who contributed the majority of generated questions, whereas students (around 7 %) also engaged in both creation and evaluation, though at a lower average volume.

System adoption was observed across all 17 academic schools. Although some schools participated modestly, others, such as Industrial Engineering, Education, and Architecture, represented several hundred generated questions each. To understand not only how much but also what kind of content was generated, we analyzed the thematic footprint of each school. Table 3 illustrates the distribution of inferred topics across the questions generated during the pilot, based on user-declared affiliations. These categories capture the disciplinary spread but also present overlaps between related areas (e.g., "Engineering" and "Engineering & Architecture"), which should be interpreted as descriptive rather than a strict taxonomy. The results indicate that engineering and architecture are more prominently represented, while less frequent categories correspond to smaller or more specialized faculties.

These results show that QuerIA was not only adopted at scale, but also engaged with flexibly across academic domains. Some users leveraged the system for subject-specific content generation, while others explored cross-cutting themes, reflecting both disciplinary needs and diverse teaching practices. The breadth of participation, both in number of users and in thematic scope, demonstrates that QuerIA was able to support heterogeneous educational contexts and instructional goals. It should be noted that topic inference relied on automatic classification, which can produce overlapping or ambiguous categories.

5.2. Difficulty calibration and linguistic consistency

To answer **RQ1**, we investigate whether the system-generated questions varied consistently across predefined difficulty levels in terms of lexical, syntactic, and cognitive characteristics. We conducted a multifaceted analysis combining linguistic, statistical, and psychometric methods. The objective was to determine whether the predefined difficulty categories-*Easy*, *Medium*, and *Difficult*-manifested in coherent textual patterns and aligned with users' performance during interaction with the system.

Across the 5987 generated questions, the distribution of difficulty levels was approximately balanced by design: 34.5 % were labeled as Easy, 35.7 % as Medium, and 29.8 % as Difficult. To verify whether these labels were reflected in the questions' linguistic structure, we calculated a series of surface-level indicators, including token count, word count, verb and noun frequency, sentence count, and a Flesch-based readability score adapted to Spanish. As shown in Table 4, questions labeled as *Difficult* were significantly longer, contained more verbs and nouns, and exhibited lower readability, confirming an expected increase in syntactic complexity.

To confirm whether these differences were statistically significant, we applied non-parametric Kruskal-Wallis tests for each linguistic variable. The results, shown in Table 5, revealed highly significant differences ($p < 0.001$) for all core metrics, including number of tokens, words, verbs, nouns, and readability. Sentence count also showed significant variation, although with smaller effect size. These results support the internal consistency of the difficulty taxonomy and its linguistic realization.

To complement this analysis, we applied a Rasch model from item response theory to evaluate whether the system's difficulty labels aligned with actual user performance. For each question, item difficulty parameters were estimated based on response correctness patterns across users, allowing for a psychometric comparison with the predefined categories.

The observed alignment was consistent with expectations: questions labeled as *Easy* were more likely to be answered correctly, while those labeled as *Difficult* showed lower success rates. Infit and outfit statistics for most items fell within acceptable ranges, indicating good model fit and supporting the internal coherence of the difficulty taxonomy. Moreover, a focused validation exercise confirmed this perception among academic users: 95 % agreed with the *Easy* label, and 79 % with the *Difficult* label.

These empirical findings are further enriched by the lexical analysis presented earlier. Together, both dimensions, linguistic structure and response-based performance, provide converging evidence that QuerIA's difficulty control mechanism produced questions that are not only syntactically and semantically differentiated, but also pedagogically valid. The system was able to implement a practical taxonomy of cognitive challenge that is both measurable and meaningful in instructional contexts.

The three-tier classification used in the pilot was defined based on the validation study conducted as part of this work, which confirmed that grouping Bloom's categories into easy, medium, and difficult levels provides a practical approximation for educational use. We note, however, that cognitive levels and perceived difficulty do not always coincide; for example, a lower-level factual question may be experienced as difficult if the content is unfamiliar. This distinction underscores the importance of contextual factors in interpreting difficulty labels, an aspect further discussed in Section 6.

5.3. Pedagogical perception and thematic variation

This subsection responds to **RQ3**, examining how users perceived the clarity, alignment, and educational value of system-generated questions, and how those perceptions vary across topics and formats. These perceptions are central to understanding whether the system produced not only syntactically plausible but also educationally relevant materials.

Feedback was collected via post-task evaluations, where users rated a selection of their generated questions along six dimensions: *Clarity*, *Complexity*, *Alignment with context*, *Overall quality*, *Pedagogical value*, and *Cognitive challenge*. Each dimension was rated on a 3-point ordinal scale, with higher values indicating stronger agreement with the respective criterion.

Table 6 presents the aggregated results across major academic domains, based on a zero-shot thematic classification of the question contexts. While most domains received consistently positive scores, subtle differences emerged. Questions associated with *Health* and *Science & Environment* obtained the highest average ratings in clarity, alignment, and pedagogical value. In contrast, those from *Engineering* and *Education* showed slightly lower averages, particularly in cognitive challenge and overall quality. The disciplinary imbalance observed in the pilot further contextualizes these findings. The predominance of engineering-related faculties suggests that the system may have been particularly well suited for technical or structured content, where factual recall and procedural reasoning are more easily captured by Bloom-aligned templates. In contrast, subjects in the humanities and social sciences may pose additional challenges, as conceptual nuances and interpretive skills are less amenable to purely text-based prompts. Future studies should aim to engage a more balanced set of disciplines to assess whether similar perceptions and performance can be observed in less structured academic contexts.

Table 3
Relative distribution of inferred topics across academic schools (percentages).

School	Education	Architecture	Topography and Geomatics	Research Support	Computer Science
School of Architecture	36	48	14	8	1
School of Aerospace Engineering	37	21	17	15	0
School of Agricultural Engineering	143	52	42	9	3
School of Civil Engineering	62	28	27	10	4
School of Building Engineering	158	62	52	8	6
School of Industrial Engineering	302	43	25	19	2
School of Computer Science	103	17	44	154	29
School of Naval Engineering	57	11	24	10	3
School of Engineering and Industrial Design	75	19	25	15	3
School of Energy and Mining Engineering	32	9	13	6	0
School of Forestry and Natural Environment	56	10	32	5	0
School of Information Systems Engineering	221	508	65	184	271
School of Telecommunication Systems Engineering	54	31	56	20	4
School of Telecommunications	9	5	21	14	5
School of Geomatics and Topography	23	2	70	6	2
Education and Innovation Center	16	6	10	43	2
Faculty of Physical Activity and Sport Sciences	23	5	7	5	0

Table 4
Average linguistic features of questions by difficulty level.

Difficulty	Tokens (mean)	Words (mean)	Verbs (mean)	Readability (mean)
Easy	11.69	9.45	0.97	96.93
Medium	13.21	11.07	0.93	91.36
Difficult	19.46	16.79	1.59	82.61

Table 5
Results of Kruskal-Wallis tests comparing linguistic features across difficulty levels.

Feature	Statistic	p-value
num_tokens	2400.65	0.00e+00
num_words	2336.85	0.00e+00
num_sentences	8.38	1.51e-02
num_nouns	1372.09	1.13e-298
num_verbs	953.42	9.26e-208
readability_score	590.13	7.15e-129

Table 6
Mean user ratings by topic group.

topic_group	clarity	complexity	alignment	pedagogical	quality
Education & Training	2.26	2.39	2.47	2.26	1.96
Engineering	2.13	2.21	2.30	2.25	1.91
Engineering & Architecture	2.48	2.27	2.58	2.32	2.18
Health	2.33	2.56	2.63	2.41	2.26
Other	2.36	2.38	2.42	2.26	2.07
STEM	2.55	2.45	2.65	2.30	2.30
Science & Environment	2.68	2.84	2.60	2.44	2.36

Overall, the mean scores suggest that users found the majority of questions to be clear, contextually appropriate, and pedagogically meaningful. Clarity and alignment ratings frequently exceeded 2.3 out of 3 across topic groups, reflecting the system's ability to generate relevant and understandable content. Pedagogical and cognitive ratings exhibited more variability, potentially reflecting discipline-specific expectations or variation in instructional goals.

These findings indicate that QuerIA was generally successful in producing educationally useful content across diverse domains. The observed variation across topics does not undermine the system's overall performance, but rather highlights the need for contextual sensitivity in instructional generation-an aspect the platform's prompt-based design is well suited to support.

In addition to thematic variation, we also examined how linguistic complexity differed between question formats. QuerIA supported two main item structures: *multiple choice* and *open-ended*. These formats serve different pedagogical purposes and may encourage distinct patterns of prompt formulation and content generation.

Table 7
Mann-Whitney *U* test results comparing linguistic features across question formats.

Feature	U Statistic	p-value
num_tokens	1086884.0	5.75e-56
num_words	1099143.0	5.74e-54
num_sentences	1741496.0	9.84e-01
num_nouns	1320634.5	2.98e-25
num_verbs	1574025.5	4.11e-06
readability_score	1897289.5	1.87e-04

To assess whether the two formats differed significantly in terms of surface-level linguistic complexity, we conducted a series of Mann-Whitney U tests. This non-parametric test was selected because it does not assume normality and is appropriate for comparing the distributions of continuous features across two independent groups.

The analysis focused on six core features: number of tokens, words, sentences, nouns, verbs, and readability. Results are shown in Table 7. Statistically significant differences ($p < 0.001$) were observed for most variables. Open-ended questions contained more tokens, words, nouns, and verbs, indicating greater syntactic complexity. Sentence count did not differ significantly, and readability scores showed only marginal variation. These patterns suggest that open-ended items tend to be more linguistically elaborate, likely due to their need for broader prompts and less constrained phrasing.

These structural differences are pedagogically meaningful. Open-ended questions typically aim to elicit analytical or constructive responses, often requiring richer lexical scaffolding. In contrast, multiple-choice items emphasize clarity and brevity, especially for effective distractor design. By supporting both formats, QuerIA enabled a flexible range of instructional designs adapted to diverse learning goals and assessment strategies.

To provide further detail on feedback activity, we analyzed who submitted evaluations and how they were distributed. Out of the 341 registered participants, 53 users (15.5 %) contributed at least one evaluation, yielding a total of 849 feedback entries. Although around 7 % of the overall participants were students, all feedback providers in this pilot were instructors, reflecting their stronger involvement in evaluating the generated content. The activity was highly concentrated: the top six providers (11.3 % of feedback users) accounted for 53.0 % of all comments, while the top twenty (37.7 %) produced 80.8 %. In terms of item type, 731 feedback entries (86.1 %) referred to multiple-choice questions (MCQs) and 118 (13.9 %) to open-ended questions (OEQs). These figures clarify that feedback came from a limited but engaged group of

Table 8
Summary of research questions, analysis dimensions, and main findings.

RQ	Focus	Main findings	Section/Table
RQ1	Difficulty calibration in generated questions	The three predefined levels (Easy, Medium, Difficult) correspond to distinct lexical and syntactic patterns; Rasch analysis confirms significant differences in difficulty estimates.	Section 5.2, Tables 5 and 4
RQ2	Adoption and user activity	341 participants across 17 schools; participation was voluntary. Activity distribution is highly skewed, with a small group of very active users. Separate statistics report usage patterns for students vs. instructors.	Section 4, Tables 1–3
RQ3	Perceived quality and pedagogical value	Multiple-choice questions (MCQs) received higher ratings in clarity and perceived usefulness. Open-ended questions (OEQs) were valued for promoting reflection, though automatic scoring requires further validation. Feedback distributions are reported by item type and topic.	Section 5.3, Table 6

instructors, with activity unevenly distributed across participants and focused primarily on MCQs.

5.4. Summary of findings

The evaluation of QuerIA across real-world usage, question generation, and user feedback revealed consistent evidence of pedagogical and technical robustness. Each dimension of analysis was guided by a specific research question and addressed through a combination of empirical metrics, statistical validation, and perceptual data.

First, we investigated how the platform was adopted across academic settings. The system was actively used throughout the semester by students and instructors from ten schools, resulting in nearly 6000 questions and over 300 questionnaires. While usage varied in intensity, participation was broad and sustained. Thematic analysis showed that most users engaged with 3-6 academic domains, and topic distributions across schools aligned well with institutional curricula, indicating that the system was adaptable to a range of disciplinary contexts.

Second, we evaluated whether QuerIA could generate questions with controlled and meaningful variation in difficulty. Results showed that the system-assigned difficulty labels (*Easy*, *Medium*, *Difficult*) corresponded to significant and consistent differences in linguistic complexity. Non-parametric tests confirmed the statistical robustness of these distinctions. Moreover, the difficulty categories aligned with user performance and perception, as validated through a Rasch analysis and targeted evaluation, suggesting that the control mechanism was pedagogically valid.

Finally, we examined how users perceived the pedagogical quality of the generated content. Feedback collected along six instructional dimensions revealed generally positive evaluations, particularly in clarity, contextual alignment, and educational value. While some variation was observed across academic domains and question formats, ratings remained consistently high across the board. Open-ended questions were found to be more lexically complex than multiple-choice items, reflecting their broader cognitive scope. Together, these findings indicate that QuerIA was able to produce content that was not only technically sound but also instructionally relevant and well-received by users.

Table 8 summarizes the evaluation results for the three research questions defined in Section 1, highlighting their analytical focus, the main findings, and the corresponding sections and tables where the results are reported. Additional analyses on thematic classification and operational implications are presented in Section 6 as complementary insights beyond the core research questions.

6. Discussion, limitations and conclusions

This section reflects on the implications of the results, acknowledges limitations, outlines directions for future work, and concludes with the main contributions of the study. By situating the findings in relation to prior research, we highlight both the pedagogical and managerial relevance of QuerIA.

The evaluation showed that QuerIA was adopted across a diverse academic landscape, with sustained usage by students and instructors from several schools. Thematic coverage reflected institutional curricula, and most users engaged with multiple domains, indicating a degree of exploratory and interdisciplinary use.

The system's difficulty control mechanism proved consistent across both linguistic and empirical dimensions. Surface-level metrics such as token and verb counts varied systematically by difficulty level, and non-parametric tests confirmed the statistical significance of these patterns. A Rasch-based validation further supported alignment between system-assigned labels and perceived or observed difficulty, suggesting that the taxonomy implemented by QuerIA is pedagogically meaningful.

User feedback highlighted consistently high ratings in clarity, contextual alignment, and overall value. These results were robust across topics and formats, with open-ended questions showing greater linguistic complexity, and thematic domains such as Health and Science receiving slightly higher scores. This suggests that the system's outputs were generally aligned with instructional expectations.

From an institutional perspective, deployment choices proved critical. Local hosting was feasible on standard university infrastructure, offering cost reduction, data governance, and independence from commercial APIs. However, sustainability and scalability remain open challenges, since keeping pace with rapidly evolving LLMs or serving larger cohorts may require additional resources. At the same time, instructors expressed concerns about sending student data to external providers, underlining that trust and governance are decisive factors for adoption. These findings connect technical feasibility with managerial considerations, showing how deployment strategies shape acceptance.

Beyond technical aspects, the study also contributes to management and educational research. By linking generative AI with Bloom's taxonomy, QuerIA illustrates how pedagogical frameworks can guide system design. At the institutional level, usage analytics and feedback data highlight how such systems may inform curriculum design and resource allocation. These implications extend the relevance of our findings beyond NLP, positioning QuerIA as a reference for integrating AI responsibly into higher education.

6.1. Ethical considerations

QuerIA was not conceived as a replacement for institutional grading systems but as a tool to support self-learning and formative practice. Its primary purpose is to provide students with opportunities to generate and answer questions aligned with their course materials, enabling practice at different speeds and levels of difficulty. This design reduces the risks associated with high-stakes or summative assessment.

Nevertheless, according to the European AI Act, educational applications of AI in assessment are categorized as "high-risk." This highlights the importance of ensuring transparency, fairness, and accountability. In QuerIA, local deployment supports data governance and privacy, while automatic evaluation of open-ended questions can be complemented with human oversight. Future iterations will strengthen hybrid validation strategies, combining AI-supported feedback with instructor or peer review to align with emerging ethical and regulatory standards.

6.2. Limitations

Despite these positive results, several limitations should be considered. A first set of constraints relates to the available data. Response-level data from students was not available for all questions, which

Table 9
Distribution of pilot participants by school (percentages over reported affiliations).

School	Percentage of Participants
School of Architecture	6.0 %
School of Aerospace Engineering	4.8 %
School of Agricultural Engineering	14.0 %
School of Civil Engineering	9.6 %
School of Building Engineering	4.8 %
School of Industrial Engineering	11.6 %
School of Computer Science	6.0 %
School of Naval Engineering	1.6 %
School of Engineering and Industrial Design	8.4 %
School of Energy and Mining Engineering	3.6 %
School of Forestry and Natural Environment	4.0 %
School of Information Systems Engineering	8.4 %
School of Telecommunication Systems Engineering	7.6 %
School of Telecommunications	4.0 %
School of Geomatics and Topography	0.4 %
Education and Innovation Center	1.2 %
Faculty of Physical Activity and Sport Sciences	4.0 %

constrained psychometric validation to aggregate-level and perception-based analysis. This limited the scope of inferences regarding learning impact. In addition, user ratings were unevenly distributed across faculties, with engineering-related schools contributing the majority of users and generated content. This self-selection bias may have influenced the observed results and limits the generalizability of the findings across all academic domains.

A second set of limitations concerns methodological aspects of the system. Thematic classification was performed using a zero-shot multilingual model, which, while scalable, may introduce errors in edge cases or interdisciplinary content. Further refinement and validation of topic annotations would strengthen future analyses. Likewise, the contribution of the semantic chunking strategy to question quality has not yet been independently validated. Although it was designed to produce topically coherent segments and mitigate the constraints of the LLM context window, its comparative effectiveness against simpler approaches remains untested.

Finally, the current version of QuerIA only supports lower and intermediate levels of Bloom's taxonomy and does not yet cover advanced categories such as "Create." This limitation reflects the difficulty of current LLMs in reliably assessing synthesis, originality, and open-ended creative outputs. Such tasks require not only linguistic fluency but also validation mechanisms that go beyond the model's internal representations, making them particularly challenging to automate. The reliability of the automated grading component for open-ended questions also remains unverified against expert annotations. While an illustrative example of model-generated feedback is provided (Fig. 3), no systematic comparison with human raters was conducted during the pilot. As a result, the validity of the scoring mechanism cannot yet be confirmed, and its outcomes should be interpreted with caution.

Third, user ratings were voluntarily submitted and unevenly distributed across domains and participants. This introduces potential response bias, with overrepresentation of more active or experienced users. Caution is advised when generalizing subjective perceptions to the full user base.

The three-level mapping of Bloom's taxonomy should be understood as a pragmatic approximation. Although validated in this study, it cannot fully capture the nuances of perceived difficulty, and advanced categories such as "Create" remain unsupported. Addressing this gap is part of our future work, which will explore strategies for generating more complex and open-ended questions.

The representativeness of the sample also poses constraints. Although 341 participants from 17 faculties were involved, participation was unevenly distributed, reflecting possible self-selection effects (Table 9). This imbalance may limit generalizability, and future studies should aim for more balanced cross-faculty participation.

Other components were not fully validated during the pilot. The contribution of semantic chunking to quality improvement was not independently tested, the reliability of automatic grading for open-ended questions needs benchmarking against expert annotations, and the diversity and potential biases of generated items remain to be systematically measured. Moreover, resilience against strategic manipulation by students (e.g., adversarial responses to influence grading) has not yet been evaluated.

Finally, ethical and regulatory aspects deserve attention. As noted, educational AI applications fall under the "high-risk" category of the European AI Act. Ensuring compliance will require additional safeguards, such as transparency of model behavior and structured mechanisms for human oversight.

6.3. Future work

Several directions for future work emerge from this evaluation. First, methodological refinements are needed to strengthen the empirical basis of QuerIA. Incorporating individual-level response data would enable more robust modeling of learning outcomes, including longitudinal tracking of user performance and fine-grained Rasch calibration. In parallel, a systematic comparative study of segmentation strategies will benchmark semantic chunking against paragraph-based and fixed-size baselines, evaluating their impact on question clarity, contextual alignment, and difficulty calibration. Such analyses will combine linguistic metrics (e.g., lexical diversity, syntactic complexity) with user-centered evaluations to quantify the added value of semantic chunking.

Second, the classification of themes and difficulty could be enhanced through semi-supervised or active learning methods, combining automated inference with expert feedback. Extending Bloom's taxonomy to higher-order categories such as "Create" represents a particularly important direction for future work. Technical strategies may include multi-step prompting pipelines that scaffold the generation process, hybrid workflows that combine LLM suggestions with expert or peer review, and integration with external knowledge sources or validation tools to ensure the originality and relevance of generated items. Such approaches would enable the system to move beyond factual, conceptual, and procedural levels, supporting more advanced forms of cognitive engagement. In addition, future studies should evaluate the reliability of the automated grading component for open-ended questions. This can be achieved through inter-rater reliability analyses, such as Cohens Kappa or Krippendorff's Alpha, comparing LLM-based scores with those provided by expert instructors for a representative subset of responses. Establishing such empirical validation will be crucial to confirm the robustness of the OEQ assessment mechanism and to support its potential adoption in real instructional settings.

Third, QuerIA could integrate user modeling components to adapt generation not only to prompt characteristics, but also to user profiles, instructional goals, or curriculum progression. This would transform the system from a static generator into an adaptive co-authoring assistant.

Finally, further research is needed on the integration of generative tools into real classroom workflows, including co-design with educators and analysis of student learning trajectories. Such studies would inform the pedagogical positioning and long-term impact of systems like QuerIA.

6.4. Conclusions

This study presented an empirical evaluation of QuerIA, a generative system for multilingual question authoring, deployed in a real-world academic environment. Through linguistic analysis, statistical validation, and user feedback, we investigated its adoption, the effectiveness of its difficulty control, and the pedagogical quality of its outputs.

Findings indicate that QuerIA is capable of producing linguistically coherent, thematically diverse, and pedagogically meaningful content at scale. Its design supports flexible and adaptive use across domains, enabling educators and students to generate assessment material that aligns with instructional needs.

A key contribution of this work is the validation of generative question design in authentic higher-education contexts, involving hundreds of instructors and students across 17 faculties. This large-scale evidence goes beyond synthetic or expert-only evaluations, offering a rare view of how such systems are received in practice.

Moreover, the demonstration of a locally deployable LLM-based system provides a concrete example of how educational institutions can adopt generative AI while maintaining control over infrastructure and data governance. QuerIA is not designed to replace instructors in their teaching or assessment duties, but to accompany and complement their work. The possibility of deploying the system locally on standard, low-resource hardware ensures maximum flexibility, allowing individual educators to potentially operate their own models adapted to their course materials and teaching style. This level of personalization and data sovereignty is difficult to achieve with large-scale commercial LLMs, further underlining the distinctive value of QuerIA in academic settings.

Finally, by linking question generation with Bloom's taxonomy and releasing an open dataset of real interactions, this work contributes to bridging educational technology with management and learning analytics research. Future developments may build on this foundation to explore advanced cognitive levels, multimodal content, and adaptive personalization, strengthening both pedagogical practice and institutional decision-making.

These results position QuerIA as a viable infrastructure for augmenting formative assessment and instructional design in higher education. The methodological framework established here may also inform future evaluations of generative educational systems.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used *WriteFull* and *GPT-4* to improve language. The authors reviewed and edited the content as needed and take full responsibility for the content.

Funding

This research was partially supported by the Universidad Politécnica de Madrid through the 2023-2024 Educational Innovation Project "Automatización de Evaluaciones: Desarrollo de Cuestionarios a través de Procesamiento de Lenguaje Natural y Aprendizaje Automático" (IE24.6109).

CRediT authorship contribution statement

Carlos Badenes-Olmedo: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing; **Paul Eyzaguirre:** Data curation, Validation, Resources, Writing – review & editing.

Data availability

The dataset and sourcecode is available at Zenodo (https://doi.org/10.5281/zenodo.16669858)

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank the Servicio de Innovación Educativa at Universidad Politécnica de Madrid for supporting the pilot evaluations in academic settings.

References

Doughty, J. A., Wan, Z., Bompelli, A., Qayum, J., Wang, T., Zhang, J., Zheng, Y., Doyle, A., Sridhar, P., Agarwal, A., Bogart, C., Keylor, E., Kultur, C., Savelka, J., & Sakr, M. F. (2023). A comparative study of AI-generated (GPT-4) and human-crafted MCQs in programming education. *Proceedings of the 26th Australasian Computing Education Conference*.

Duong-Trung, N., Wang, X., & Kravčík, M. (2024). BloomLLM: Large language models based question generation combining supervised fine-tuning and bloom's taxonomy. In *Technology enhanced learning for inclusive and equitable quality education* (pp. 93–98). Springer Nature Switzerland.

Gnanasekaran, D., Kothandaraman, R., & Kaliyan, K. (2021). An automatic question generation system using rule-based approach in bloom's taxonomy. *Recent Advances in Computer Science and Communications*, *14*(5), 1477–1487. https://doi.org/https://doi.org/10.2174/2213275912666191113143335

Guo, Hongwen, Johnson, Matthew S., Ercikan, Kadriye, Saldivia, Luis, Worthington, Michelle (2024). Large-scale assessments for learning: A human-centred AI approach to contextualizing test performance. *Journal of Learning Analytics*. https://doi.org/10.18608/jla.2024.8007

Han, X., & Li, K. (2017). Hierarchical-dynamic embedding for zero-shot object recognition. In *2017 international conference on computational science and computational intelligence (CSCI)* (pp. 520–525). IEEE Computer Society. https://doi.org/10.1109/CSCI.2017.88

Hu, J., & Yang, B. (2024). Posts quality prediction for stackoverflow website. *IEEE Access*, *12*, 135601–135615. https://doi.org/10.1109/ACCESS.2024.3440879

Huang, Qian, Lv, Chunlan, Lu, Li, Tu, Shuang (2025). Evaluating the quality of AI-generated digital educational resources for university teaching and learning. *Systems*, *13*(3). https://doi.org/10.3390/systems13030174

Hwang, K., Wang, K., Alomair, M., Choa, F.-S., & Chen, L. K. (2024). Towards automated multiple choice question generation and evaluation: Aligning with bloom's taxonomy. In *Artificial intelligence in education* (pp. 389–396). Springer Nature Switzerland.

Jauhiainen, J. S., & Guerra, A. G. (2025). Generative AI in education: ChatGPT-4 in evaluating students' written responses. *Innovations in Education and Teaching International*, *62*(4), 1377–1394. https://doi.org/10.1080/14703297.2024.2422337

Khalil, Mohammad, Liu, Qinyi, Jovanovic, Jelena (2025). Ai for data generation in education: Towards learning and teaching support at scale. *British Journal of Educational Technology*, *56*(3), 993–998. https://doi.org/10.1111/bjet.13580

Li, F., & Yeh, M.-C. (2021). Generalized zero-shot recognition through image-guided semantic classification. In *2021 IEEE international conference on image processing (ICIP)* (pp. 2483–2487). https://doi.org/10.1109/ICIP42928.2021.9506245

Li, X., & Guo, Y. (2015). Max-margin zero-shot learning for multi-class classification. In *Proceedings of the eighteenth international conference on artificial intelligence and statistics* (pp. 626–634). PMLR (*vol. 38*). Proceedings of Machine Learning Research.

Li, X., Guo, Y., & Schuurmans, D. (2015a). Semi-supervised zero-shot classification with label representation learning. In *2015 IEEE international conference on computer vision (ICCV)* (pp. 4211–4219). https://doi.org/10.1109/ICCV.2015.479

Li, X., Liao, S., Lan, W., Du, X., & Yang, G. (2015b). Zero-shot image tagging by hierarchical semantic embedding. In *Proceedings of the 38th international ACM SIGIR conference on research and development in information retrieval* SIGIR '15 (p. 879–882). Association for Computing Machinery. https://doi.org/10.1145/2766462.2767773

Li, Z., Cukurova, M., & Bulathwela, S. (2025). A novel approach to scalable and automatic topic-controlled question generation in education. In *Proceedings of the 15th international learning analytics and knowledge conference* LAK '25 (p. 148–158). Association for Computing Machinery. https://doi.org/10.1145/3706468.3706487

Lodovico Molina, I., Švábenský, V., Minematsu, T., Chen, L., Okubo, F., & Shimada, A. (2024). Comparison of large language models for generating contextually relevant questions. In *Technology enhanced learning for inclusive and equitable quality education: 19th European conference on technology enhanced learning, EC-TEL 2024, Krems, Austria, September 16-20, 2024, proceedings, part II* (p. 137–143). Springer-Verlag. https://doi.org/10.1007/978-3-031-72312-4_18

Martel, F., & Zouaq, A. (2021). Taxonomy extraction using knowledge graph embeddings and hierarchical clustering. In *Proceedings of the 36th annual ACM symposium on applied computing* SAC '21 (p. 836–844). Association for Computing Machinery. https://doi.org/10.1145/3412841.3441959

Mi, Z., & Li, K. (2024). A comparative analysis of different large language models in evaluating student-generated questions. In *2024 13th international conference on educational and information technology (ICEIT)* (pp. 24–29). https://doi.org/10.1109/ICEIT61397.2024.10540914

Owan, Valentine Joseph, Abang, Kingsley Bekom, Idika, Delight Omoji, Etta, Eugene Onor, Bassey, Bassey Asuquo (2023). Exploring the potential of artificial intelligence tools in educational measurement and assessment. *Eurasia Journal of Mathematics, Science and Technology Education*, *19*(10), 1–15. https://doi.org/10.29333/ejmste/13428

Rachid Karra, A. L. (2023). Impact of data quality on question answering system performances. *Intelligent Automation & Soft Computing*, *35*(1), 335–349.

Rithish, H., Deepak, G., & Santhanavijayan, A. (2021). Automated assessment of question quality on online community forums. In *Digital technologies and applications* (pp. 791–800). Springer International Publishing.

Shoaib, M., Sayed, N., Singh, J., Shafi, J., Khan, S., & Ali, F. (2024). Ai student success predictor: Enhancing personalized learning in campus management systems. *Computers in Human Behavior*, *158*, 108301. https://doi.org/10.1016/j.chb.2024.108301

Singh, M., Patvardhan, C., & Lakshmi, C. V. (2023). Does chatGPT spell the end of automatic question generation research? *2023 IEEE International Conference on Computer Vision and Machine Intelligence (CVMI)*, (pp. 1–6).

Toba, H., Yudha, L. G. O. P., Karnalim, O., Bunyamin, H., & Tada, T. (2025). A large language model question generator based on bloom's taxonomy template. In *Proceeding of the 2024 8th international conference on education and e-learning* ICEEL '24 (p. 25–31). Association for Computing Machinery. https://doi.org/10.1145/3719487.3719537

Wijanarko, B. D., Heryadi, Y., Toba, H., & Budiharto, W. (2021). Question generation model based on key-phrase, context-free grammar, and bloom's taxonomy. *Education and Information Technologies*, *26*(2), 2207–2223. https://doi.org/10.1007/s10639-020-10356-4

Williams, P. (2023). Ai, analytics and a new assessment model for universities. *Education Sciences*, *13*(10). https://doi.org/10.3390/educsci13101040